

Cloud Vulnerability Scanner

Agentic AI Scanner

1. Orchestration Agent

- A. When user types “Scan my AWS account” in the UI, the requests logged, the agent decodes the message and decides which cloud provider is to be scanned (AWS in this case).
- B. Returns a JSON object with keys (provider, account ID)
- C. Additional check if there are any mistakes (Spelling errors, capitalisation etc)
- D. For each provider in the JSON:
 - AI plans and triggers cloud scans (cloud resource discovery, misconfiguration check, deep scan if requested by user)
- E. Store result in database,

2. Risk Analysis Agent

- A. Receives ScanResult, generates a narrative security analysis (posture, top risks, attack paths, compliance concerns, 24-hour actions).
- B. Locally computes a risk score per finding (severity + public exposure + IAM/keys), sorts, and returns the top 20 highest-risk issues.
- C. Builds an executive summary with total resources, findings, severity breakdown, public resources, a 0–100 security score, and posture label (EXCELLENT → CRITICAL).

3. Agent Chat

- A. Generic Cloud Security Advisor: ChatGPT-like advisor for cloud security.
- B. Scan-Aware, Data-Grounded Agent: Smart explainer that understands your actual scans.

Plugins

An *adapter* that teaches the scanner how to scan a specific cloud.
Each plugin must implement **the same set of functions**.
This makes every plugin behave consistently.

These functions are the rules (the “contract”):

1. `get_provider_name()`
→ “I am AWS” or “I am GCP”
2. `discover_resources(account_id)`
→ “Here are all the resources in this cloud”
3. `check_config(resources)`
→ “Here are configuration mistakes (like public buckets)”
4. `assess_vulnerabilities(resources)`
→ “Here are security risks (like no encryption)”
5. `full_scan(account_id)`
→ Runs all the above steps and returns results

Every plugin—AWS, GCP, OpenAI—implements these same methods.

Plugins allow your main scanner to support multiple clouds by using small, swappable adapter modules that all follow the same interface.